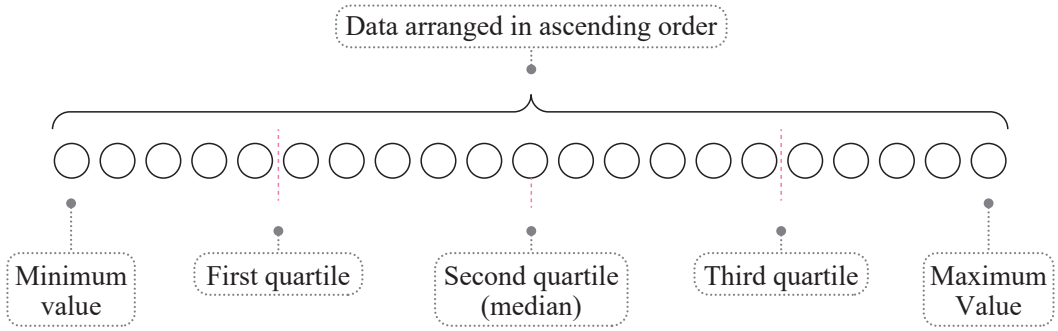
10-3 Data Analysis [2]

Point!

1 Quartiles

(1) ([1]**Quartile**) : The numbers that divide the data into four equal parts when arranged in ascending order. Quartiles are referred to as follows in ascending order: ([2]**first quartile**), ([3]**second quartile**), ([4]**third quartile**). The second quartile is the ([5]**median**).



(2) Steps for determining the quartiles

[1] Arrange the data in ascending order and determine the ([6]**median**).

[2] With the median in [1] as the boundary, divide into two groups: ([7]**less than or equal to the median**) and ([8]**greater than or equal to the median**).

[3] For each group created in [2], determine the ([9]**median**).

2 Box-and-whisker Plots

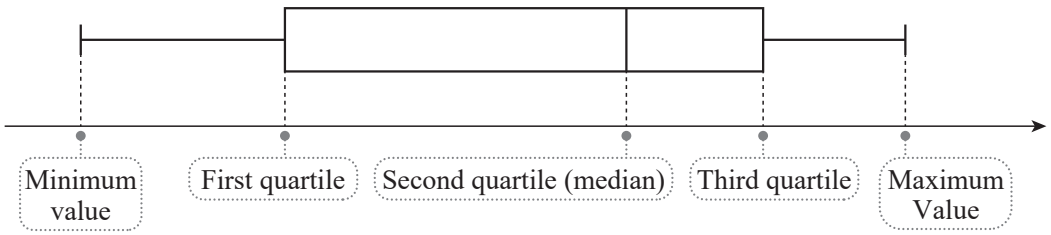
(1) ([10]**Box-and-whisker plot**) : A figure using a box and lines (whiskers) to represent the minimum value, first quartile, second quartile, third quartile, and maximum value in a single figure, as shown below.

(2) Steps to draw a box-and-whisker plot

[1] Determine the ([11]**minimum value**), ([12]**maximum value**), and ([13]**quartiles**), and draw five vertical lines representing the points.

[2] Draw a box with the first quartile as the left edge and the third quartile as the right edge.

[3] Draw a horizontal line from each end of the box to the minimum and maximum values.



*A box-and-whisker plot can also be drawn vertically.

Warm Up

Answer the following questions.

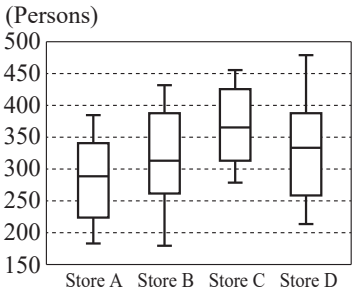
(1) The following data represents the math test scores of ten students.

58 72 52 61 36 43 20 32 38 47 (Points)

[1] Provide the first quartile, second quartile, and third quartile.

[2] Draw a box-and-whisker plot based on this data.

(2) The figure on the right shows data on the number of daily customers at stores A, B, C, and D over a span of 31 days in a box-and-whisker plot.



[1] List all stores with days where the number of customers exceeded 450.

[2] Provide the minimum number of days where the number of customers exceeded 300 at store B.

[3] List all stores that had eight or more days with fewer than 250 customers.

Explanation

(1) [1] When you arrange the data in ascending order,

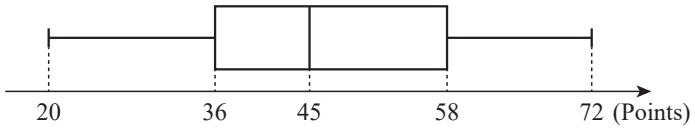
20, 32, 36, 38, 43, 47, 52, 58, 61, 72

The median is (43 + 47) ÷ 2 = 45 (points).

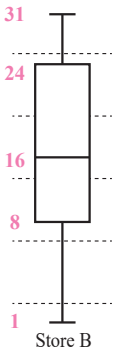
The median of the group less than or equal to the overall median, 20, 32, 36, 38, 43, is 36. The median of the group greater than or equal to the overall median, 47, 52, 58, 61, 72, is 58.

Therefore, the first quartile is 36 points, the second quartile is 45 points, and the third quartile is 58 points.

[2] The minimum value is 20, the maximum value is 72, the first quartile is 36, the second quartile is 45, and the third quartile is 58, so the box-and-whisker plot will look like the figure below.



(2) Since the data was collected over 31 days, the median is the 16th value, the first quartile is the 8th value, and the third quartile is the 24th value. Accordingly, write it into the figure as shown on the right.



[1] Choose the box-and-whisker plots where the maximum exceeds 450, which would be store C and store D.

[2] From the box-and-whisker plot of store B, it can be determined that the median exceeds 300 people. From the figure on the right, the median corresponds to the value of the 16th data point when counted from smallest to largest. Therefore, from the 16th to the 31st data points, the number exceeds 300 people, so it is 16 days.

[3] From the figure on the right, the first quartile is the 8th data point when counted from the smallest. Therefore, we should select the box-and-whisker plot where the first quartile is less than 250 people, so store A.

Try

1 The following data represents the test scores of ten individuals. Answer the following questions.

68 90 75 64 72 32 46 86 59 82 (Points)

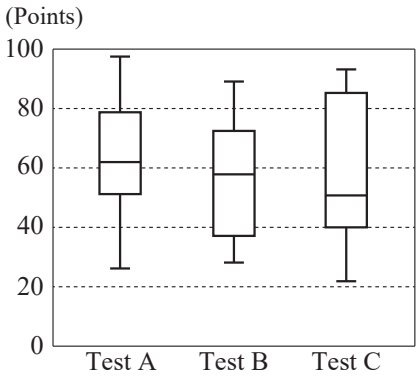
(1) Provide the first quartile, second quartile, and third quartile.

(2) Draw a box-and-whisker plot based on this data.

2 The figure on the right is a box-and-whisker plot representing the scores of tests A, B, and C taken by 50 first-year high school students. Answer the following questions.

(1) Identify all tests where there are 13 or more students scoring 80 or above.

(2) Identify all tests where more than half of the students scored below 60 points.



Exercise

1 The following data represents the Japanese language test scores of 10 individuals. Answer the following questions.

66 77 79 46 30 25 54 95 50 86 (Points)

(1) Provide the first quartile, second quartile, and third quartile.

(2) Draw a box-and-whisker plot based on this data.

2 The figure on the right is a box-and-whisker plot representing the scores of tests A, B, and C taken by a class of 30 high school students.

Answer the following questions.

(1) Identify all tests where there are 8 or more students scoring 80 or above.

(2) Identify all tests where more than half of the students scored below 60 points.

